EQUIPMENT CLEANING VALIDATION PROTOCOL

Product Name: Updated Product NEW

Product Code: TPNEW01

Location: Indore

Manufacturing Block: Plant-2

Batch Size: 150.0 ± 100.0 Kg

Date of Issue: May 2026

Protocol No.: PRO/24/079/R01 (Revision R00)

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1. PROTOCOL APPROVAL

Activity	Department	Signature	Name	Date
Prepared By	Quality Assurance	____________	Devendra Bairagi	19/05/2026
Checked By	Production	____________	__________	__________
Checked By	Powder Processing Area	____________	__________	__________
Checked By	Engineering	____________	__________	__________
Checked By	Quality Control	____________	__________	__________
Checked By	Quality Assurance	____________	__________	__________
Approved By	Head Quality Assurance	____________	__________	__________

2. INTRODUCTION

Cleaning is necessary to avoid cross contamination from previous active ingredients, raw materials, intermediates, residual solvents to subsequent products. Contamination from lubricant used for equipment, cleaning agent (if any) and cleaning tools such as brushes, mops etc. used during cleaning process. It also establishes the bio-burden load of equipment after completion of cleaning process. Present cleaning validation study involves evaluation of cleaning procedure established for cleaning of various equipment used in the manufacturing of Updated Product NEW through rinse / swab sampling (for micro & chemical) technique.

API product "Updated Product NEW" is planned to be manufactured in Plant-2 at manufacturing facility. Hence as per SOP, cleaning validation shall be performed in API stage from crystallization onwards by establishing maximum allowable carryover (MACO) criteria.

Regulatory Basis for this Protocol:

• ICH Q7 (GMP for Active Pharmaceutical Ingredients) - Section 5.2

• EU GMP Annex 15 (Qualification and Validation) - Section 10

• WHO TRS 1019 (Validation Guidelines) - Appendix 3 & 7

• APIC Cleaning Validation Guide 2021 (Sections 4, 5, 7, 8)

• PIC/S PI 006-3 (Cleaning Validation)

• EMA Guideline on Setting Health-Based Exposure Limits

3. OBJECTIVE

The objective of this protocol is to provide documented evidence through the scientific data to show that the cleaning procedure established for cleaning of equipment used in the manufacturing process is effective and consistently performs as expected and produces a result that meets predetermined acceptance criteria when cleaning is performed.

4. SCOPE

The scope of the protocol is to execute the cleaning validation study for the equipment used in the manufacturing process of Updated Product NEW in manufacturing area.

5. RESPONSIBILITIES

Department	Responsibilities
Quality Assurance (Team Leader)	To prepare & approve the cleaning validation protocol, coordinate the entire study activity, MACO calculatio
Quality Control	To analyze the cleaning validation samples (Chemical & Microbial), prepare specification, take swab sampl
Production	To provide information required for MACO, execute cleaning of equipment, maintain cleaning logbook, infor
Engineering	To provide complete details (Capacity & Surface area etc.) of each equipment installed.

6. CLEANING PROCEDURE

6.1 CLEANING AGENT SELECTION CRITERIA

Selection criteria of solvent for cleaning are based on the solubility pattern of Updated Product NEW into cleaning agent. Because Updated Product NEW is freely soluble in water, water is non-toxic, leaves no harmful residues, is readily available and cost-effective, and is compatible with all equipment MOCs (SS 316L).

6.2 SOLUBILITY PATTERN AND CLEANING AGENT

NAME OF PRODUCT	SOLUBILITY PROFILE
Updated Product NEW (TPNEW01)	Freely Soluble
Cleaning Agent	Purified Water
Quality Standard	IP/Ph.Eur./USP
Conductivity	NMT 1.3 µS/cm at 25°C
TOC	NMT 0.5 mg/L
Bioburden	NMT 100 CFU/mL
Temperature for Wash Cycle	40-60°C
Temperature for Final Rinse	Ambient (25 ± 5°C)

6.3 DETAILS OF EQUIPMENT USED AND CONTACT SURFACE AREA

Sr. No.	Equipment Name	Equipment No.	Capacity	Surface Area (m^2)
1	Updated Reactor NEW	TRNEW01	1200.0	30.50
			TOTAL:	30.50

6.7 DIRTY HOLD TIME (DHT) VALIDATION

Definition: Time from end of manufacturing (batch completion) to start of wet cleaning. Because residues may dry, harden, or degrade over time, becoming more difficult to remove. Maximum claimed DHT: 24.0 hours.

Time Point	Action	Sampling
T0 (Baseline)	Clean immediately after manufacturing	Swab + Rinse + Bioburden
T1 (24.0 hours)	Keep dirty for 24.0 hours, then clean	Swab + Rinse + Bioburden
T2 (48.0 hours)	Keep dirty for 48.0 hours, then clean	Swab + Rinse + Bioburden
T3 (72.0 hours)	Keep dirty for 72.0 hours, then clean	Swab + Rinse + Bioburden

6.8 CLEAN HOLD TIME (CHT) VALIDATION

Definition: Time from completion of cleaning to next use of equipment. Because cleaned equipment may accumulate dust, moisture, or microbial growth during storage. Maximum claimed CHT: 14 days.

Time Point	Action	Sampling
T0 (Day 0)	Clean equipment, sample immediately	Visual + Swab + Bioburden
T1 (Day 7)	Store for 7 days, sample	Visual + Swab + Bioburden
T2 (Day 14)	Store for 14 days, sample	Visual + Swab + Bioburden
T3 (Day 21)	Store for 21 days, sample	Visual + Swab + Bioburden

6.9 CAMPAIGN LENGTH VALIDATION

Definition: Number of consecutive batches of same product manufactured before full cleaning. Maximum Campaign Length: 10 batches or 30 days, whichever is earlier. Because residue levels remain within limits.

7. ESTABLISHMENT OF MAXIMUM ALLOWABLE LIMIT

7.1 MACO/MSC CALCULATION WITH WORST-CASE RATIONALE

Method	Formula	MACO (mg)
ADE/PDE Method	MACO = (ADE × MBS_next) / TDD_next	375000.00
TDD Method	MACO = (TDD_prev × MBS_next) / (1000 × TDD_next)	81818.18
10 ppm Method	MACO = 0.001% × MBS_next	1500.00
Selected MACO		1500.00 mg

7.2 CALCULATION OF RINSE LIMIT

Rinse Limit per Equipment (mg) = (MACO × Equipment Surface Area) / Total Surface Area

Equipment	Surface Area (m^2)	Rinse Limit (mg/equipment)
Updated Reactor NEW	30.50	1500.00

7.3 CALCULATION OF SWAB LIMIT

Swab Limit (mg/swab of 0.01 m^2) = (MACO × 0.01 m^2) / Total Surface Area = (1500.00 mg × 0.01) / 30.50 m^2 = 0.491803 mg/swab ≈ 0.18 mg/swab

7.4 STRATIFIED (STAGED) CARRYOVER CALCULATION

Total Carryover (mg) = Σ (MaxSwab■ × ESA■ / SSA■)

Equipment	ESA (dm^2)	Swab Limit (mg)	Max Contribution (mg)
Updated Reactor NEW	3050	0.491803	1500.00

8. SAMPLING PLAN

8.1 FMEA-BASED SCIENTIFIC RATIONALE FOR SELECTING SWAB SAMPLING POINTS

Risk Priority Number (RPN) Formula: RPN = Severity (S) × Occurrence (O) × Detection (D)

Equipment	Failure Mode	S	O	D	RPN	Sampling Point
SSR-311E	Residue at upper dome near shaft	8	4	3	96	■ Yes
SSR-311E	Residue near inlet valve	7	5	3	105	■ Yes
SSANFD-311A	Residue at sieve corner	9	6	3	162	■ Yes
SSANFD-311A	Residue on pressing rod	8	5	4	160	■ Yes
SSMM-311A	Residue on mesh	7	5	4	140	■ Yes
SSMM-311A	Residue on blades	8	4	3	96	■ Yes
SSSFT-311B	Residue on sieve surface	7	4	4	112	■ Yes
SSACM-311A	Residue in screw feeder	8	5	4	160	■ Yes
SS Containers	Residue on inner wall corners	5	6	4	120	■ Yes

8.3 RECOVERY STUDY AND CORRECTION FACTOR

No recovery studies found. Please add recovery data for each MOC.

9. ACCEPTANCE CRITERIA

9.1 VISUAL INSPECTION

Equipment surfaces including 'Hard to clean' areas must appear visually clean with no traces of product or other extraneous matter. Lighting: Minimum 200 lux. Distance: <30 cm.

9.2 SWAB SAMPLES (CHEMICAL)

Parameter	Acceptance Criteria	Basis
Uncorrected Result	NMT 0.491803 mg/swab	Based on MACO calculation
Recovery Correction	Apply recovery % from recovery study	As per recovery study
Corrected Result (Final)	NMT 0.491803 mg/swab	Pass/fail based on corrected result

9.3 MICROBIAL CONTAMINATION

Test	Acceptance Limit	Rationale
Total Viable Aerobic Count	NMT 100 cfu/swab of 25 cm^2	Purified water specification
Total Yeast & Molds Count	NMT 10 cfu/swab of 25 cm^2	Indicates moisture issues
Pathogens	Must be Absent	Patient safety risk

10. ANALYTICAL METHOD VALIDATION

10.2 CHEMICAL METHOD VALIDATION

Validation Parameter	Result	Acceptance Criteria	Status
LOD (Limit of Detection)	0.1663 ppm	Signal:Noise ≥ 3:1	■ Pass
LOQ (Limit of Quantitation)	0.5040 ppm	Signal:Noise ≥ 10:1	■ Pass
LOQ vs Limit Ratio	28%	≤50% required	■ Pass
Linearity (R^2)	0.999	NLT 0.990	■ Pass
Accuracy (Recovery)	98-102%	90-110%	■ Pass
Precision (RSD)	1.2%	NMT 2.0%	■ Pass
Specificity	No interference	No peaks at RT	■ Pass

11. OPERATOR QUALIFICATION FOR CLEANING

No operator qualification records found.

12. CONTINUED PROCESS VERIFICATION (PHASE 3)

Parameter	Frequency	Acceptance	Responsible
Visual Inspection	Every batch	Must be clean	Production
Rinse Sample (Conductivity)	Every batch	NMT 1.3 µS/cm at 25°C	QC
Rinse Sample (Chemical)	Every 10th batch	NMT individual equipment limit	QC
Swab Sample (Chemical)	Every 10th batch	NMT 0.18 mg/swab	QC
Bioburden Swab	Every 5th batch	NMT 100 CFU/swab	QC

13. NITROSAMINES RISK ASSESSMENT

Nitrosamine risk assessment for Updated Product NEW shows LOW to MEDIUM risk. No nitrosating agents used in process. Water nitrite controlled.

14. DATA INTEGRITY (ALCOA+) COMPLIANCE

Principle	Requirement	Compliance Method
Attributable	Who performed the action?	Electronic signatures with user ID
Legible	Can data be read?	Electronic records with clear font
Contemporaneous	Recorded at time of action?	Real-time entry, no back-dating
Original	Is this the first capture?	Raw data preserved, no re-recording
Accurate	Free from errors?	Second-person verification
Complete	All data included?	No selective reporting
Consistent	Logical sequence?	Audit trail, time stamps
Enduring	Preserved long-term?	Electronic backup, secure storage
Available	Accessible for inspection?	15-minute retrieval, organized filing

15. DETAIL OF DEVIATION / DISCREPANCIES

Details of deviation / discrepancies initiated / reported during the execution of validation study shall be reported in cleaning validation report.

16. REVALIDATION CRITERIA

#	Trigger	Action Required
1	Change in cleaning procedure	Full revalidation (3 consecutive batches)
2	Change in cleaning agent	Full revalidation + new recovery studies
3	Change in minimum batch size (decrease >20%)	Recalculate MACO; revalidate if needed
4	Major change in processing equipment	Re-qualification + revalidation
5	New product with lower ADE/PDE (≤50 µg)	Recalculate MACO; revalidate
6	Periodic revalidation	Once in 5 years

17. RESULTS

Results of three cleaning campaign batches shall be tabulated as per the enclosed format.

18. CONCLUSION

Cleaning validation report shall be prepared & conclusion shall be drawn based on the obtained results.

19. SUMMARY REPORT AND APPROVAL

Abstract shall be written on completed protocol study and results shall be tabulated.

FORM – A (Updated with Recovery Correction)

(A) Table for Chemical contamination level by Swab (with Recovery Correction), Visual Inspection & Residual rinse content equipment results:

o. No.	Location	Visual Inspection	Uncorrected (mg/swab)	Recovery %	Corrected (mg/swab)	Pass/Fail	Rinse Result (mg/Equip.)	Limit (mg/
EW01	_______	___	______	___	______	___	______	___

FORM – B (Microbiological)

(B) Table for Microbiological contamination level by swab:

Sr. No.	Equip. No.	Location	TVC (CFU/swab)	Yeast/Mold (CFU/swab)	Pathogens	Pass/Fail
1	TRNEW01	________	_______	_______	________	___

FORM – C (Visual Inspection)

(C) Table for Visual Inspection of Equipment:

Sr. No.	Equipment Name	Equipment No.	Visual Inspection Result	Inspected By	Date
1	Updated Reactor NEW	TRNEW01	________	________	________